

AI LMS Ecosystem - Executive Project Report

System Architecture • Multimodal AI Engines • Engineering RCA • Growth Roadmap

1. Executive Summary

The **AI LMS (Intelligence Learning Network)** is an ultra-modern, multimodal AI educational platform designed to automate lesson synthesis, multilingual voiceovers, video storyboards, OCR notebook digitizing, dynamic test creation with database telemetry, and adaptive roadmaps.

2. 9 AI Intelligence Modules Architecture

Module	Functionality & Technology Stack	Output Artifacts
1. Lesson Generator	Grade-tailored lesson notes + Google Imagen 3 visuals	Lesson PDF Notes
2. Audio & Video Studio	ElevenLabs TTS + Hugging Face Video + Gamma 16:9 PPT	MP3 Audio, MP4 Video, PPTX
3. OCR Converter	Handwritten notes OCR via PyMuPDF & Tesseract	Extracted Lesson Notes
4. Quiz Creator	Student profile validation + auto score grading	Worksheet & Graded PDF
5. Doubt Chatbot	24/7 conversational tutor via Gemini 2.5 Flash	Real-time Chat Answers
6. Assignment Evaluator	Auto grading subjective student answers against rubrics	Numeric Scores & Feedback
7. Learning Roadmap	Personalized 14-day study skill reinforcer	Adaptive Study Roadmap
8. Progress Analytics	Neon PostgreSQL live telemetry + Recharts visualization	Progress Telemetry PDF
9. AI Study Planner	Daily timetable revision scheduler	Personalized Timetable

3. Technical Challenges & Engineering Solutions (RCA)

Challenge	Root Cause & Symptom	Engineering Resolution
PyMuPDF Import Error	Missing native C-binary fitz in venv	Installed PyMuPDF + pypdf dual fallback in converter.py
DB Column Mismatch	Missing student_name & roll_no columns	Migrated SQLAlchemy schema and recreated Neon DB ta
HF Video Model Timeout	Queue delays on free HF endpoints	Multi-endpoint loop + automated moviepy slide MP4 falla
Indic Script Render	Pillow bitmap font missing Unicode glyphs	Dynamic Windows system font loading (Nirmala, Raavi, M
Plain Presentation	Generic python-pptx default layout	Re-engineered 16:9 Gamma-style dark theme cards with

4. Future Growth & Strategic Roadmap

- **Real-time Voice Classroom:** Upgrade Doubt Chatbot to bidirectional voice streaming using Gemini Live API.
- **School Admin Dashboard:** Class-wide score telemetry, teacher gradebook sync, and bulk assignment dispatch.
- **Offline Mobile App:** Native React Native app with local lesson caching for offline student revision.